

Service Desk Backlog — Initial Release

Service Classes reminder:

- **Expedite:** Urgent, bypass WIP limits, entire team swarms
- **Standard:** Normal priority, respect WIP limits
- **Fixed Date:** External deadline — calculate start time backward
- **Intangible:** Process improvements, technical debt (aim for 15–20% of throughput)

Sizes: Small (15–30 min) · Medium (30–60 min) · Large (60–90 min)

New Items (9 items)

Bugs (5 items)

B-01: Ticket search returns incorrect results for date-based searches

- **Type:** Bug
 - **Service Class:** Standard
 - **Size:** Small
 - **Description:** When users search tickets using date ranges or created date filters, the search function returns tickets outside the specified range. The `search_tickets()` function doesn't currently support date-based filtering, only ID and text search.
 - **Steps to reproduce:**
 1. Create tickets with different creation dates
 2. Attempt to search for tickets created in last 7 days
 3. Function fails or returns irrelevant results
 - **Expected behavior:** Search should support date range filtering (e.g., "created in last 7 days", "between date X and Y")
 - **Acceptance Criteria:**
 - ☐ Add date range parameters to `search_tickets()` function
 - ☐ Filter tickets by `created_at` timestamp
 - ☐ User can search "last 7 days", "last 30 days", or custom date range
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B-02: Closed tickets appear in "Open Tickets" view

- **Type:** Bug
 - **Service Class:** Standard
 - **Size:** Small
 - **Description:** Filter logic in `view_tickets(filter_status='open')` doesn't correctly exclude closed tickets. Some closed tickets still display in the open tickets list, confusing users about actual open ticket count.
 - **Impact:** Support staff waste time reviewing tickets that are already resolved. Metrics (# of open tickets) are incorrect.
 - **Root Cause:** Likely issue with status comparison (case sensitivity or whitespace in status field)
 - **Acceptance Criteria:**
 - ☐ `view_tickets(filter_status='open')` shows only tickets with status = 'Open'
 - ☐ Closed tickets do not appear in filtered view
 - ☐ Test with tickets in all three statuses (Open, In Progress, Closed)
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B-03: Ticket comments display in wrong order

- **Type:** Bug
 - **Service Class:** Standard
 - **Size:** Small
 - **Description:** Comments are displayed in chronological order (oldest first), but users expect reverse chronological (newest first) for easier reading of latest updates.
 - **Expected behavior:** Most recent comments should appear at the top of the comments section
 - **Acceptance Criteria:**
 - ☐ Reverse order of comments in `view_ticket_details()` function
 - ☐ Add timestamp to each comment for clarity
 - ☐ Most recent comment appears first
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B-04: Assign ticket doesn't validate staff name

- **Type:** Bug
 - **Service Class:** Standard
 - **Size:** Small
 - **Description:** `assign_ticket()` function accepts any string as staff name, including empty strings, numbers, or nonsensical input. No validation against known staff list.
 - **Impact:** Tickets get assigned to non-existent staff, causing confusion and lost tickets
 - **Acceptance Criteria:**
 - ☐ Validate staff name is not empty
 - ☐ (Optional) Maintain list of valid staff names and validate against it
 - ☐ Show error message for invalid staff names
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B-05: Search by ticket ID fails for single-digit IDs

- **Type:** Bug
 - **Service Class:** Standard
 - **Size:** Small
 - **Description:** `search_tickets()` function uses `query.isdigit()` to detect ID searches, but formatting issue causes single-digit IDs (1–9) to not be found.
 - **Steps to reproduce:**
 1. Search for ticket ID "3"
 2. Function returns "not found" even though ticket #3 exists
 - **Expected behavior:** All ticket IDs should be searchable, including single-digit IDs
 - **Acceptance Criteria:**
 - ☐ Fix ID search logic to handle single-digit IDs
 - ☐ Test searching for IDs 1–9
 - ☐ Verify search works for double-digit and triple-digit IDs as well
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Features (2 items)

F-01: Add priority field to tickets

- **Type:** Feature
- **Service Class:** Standard

- **Size:** Medium
 - **Description:** Tickets should have priority field (Low, Medium, High) to help support staff prioritize their work. Currently all tickets treated equally, but some issues are more urgent than others.
 - **Business Value:** Support staff can focus on high-priority issues first, improving customer satisfaction
 - **Acceptance Criteria:**
 - ☐ Add `priority` field to ticket dictionary (values: 'Low', 'Medium', 'High')
 - ☐ User can select priority when creating ticket (default: 'Medium')
 - ☐ Priority displayed in ticket list view (color-coded if possible: red=High, yellow=Medium, green=Low)
 - ☐ Priority displayed in ticket details view
 - ☐ (Optional) Can filter tickets by priority
 - ☐ (Optional) Can sort tickets by priority
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F-03: Add ticket categories

- **Type:** Feature
 - **Service Class:** Standard
 - **Size:** Medium
 - **Description:** Categorize tickets by type: Hardware, Software, Network, Access/Permissions, Other. Helps with reporting, metrics, and routing tickets to appropriate support specialists.
 - **Business Value:** Better ticket organization, faster resolution (route to specialists), useful metrics (which category has most tickets?)
 - **Acceptance Criteria:**
 - ☐ Add `category` field to tickets
 - ☐ User can select category when creating ticket (dropdown/menu with 5 options)
 - ☐ Category displayed in ticket list view
 - ☐ Can filter tickets by category
 - ☐ Default category: 'Other'
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Improvements (2 items)

I-01: Refactor ticket listing for better performance

- **Type:** Improvement
 - **Service Class:** Intangible
 - **Size:** Medium
 - **Description:** Current implementation of `view_tickets()` loads all tickets into memory and iterates through the entire list every time, even when filtering. For large ticket volumes (1000+ tickets), this becomes slow. Optimize using generators or more efficient filtering.
 - **Technical Benefit:** Faster performance, better scalability, more professional code
 - **Why Intangible:** No user-facing feature change, just internal code quality improvement
 - **Acceptance Criteria:**
 - ☐ Refactor `view_tickets()` to use generator expressions or list comprehensions
 - ☐ Implement pagination (show 20 tickets per page, "Next" to see more)
 - ☐ Test with large dataset (100+ tickets) to verify performance improvement
 - ☐ No change in user-visible behavior (same output, just faster)
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I-03: Add input validation and error handling

- **Type:** Improvement
- **Service Class:** Intangible

- **Size:** Medium
 - **Description:** Current code has minimal error handling. Add try-except blocks, validate user inputs, provide helpful error messages. Currently crashes or behaves unexpectedly on invalid input.
 - **Technical Benefit:** More robust system, better user experience (helpful error messages), prevents crashes
 - **Examples of Missing Validation:**
 - Non-integer input when ticket ID expected → ValueError crash
 - Empty string input for required fields → silent failure or bad data
 - Non-existent ticket ID → generic error message
 - **Acceptance Criteria:**
 - ☐ Add try-except blocks around all user input parsing
 - ☐ Validate all required fields are non-empty
 - ☐ Provide specific error messages (not generic "Error occurred")
 - ☐ Graceful handling of edge cases (e.g., closing already-closed ticket)
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Summary

#	ID	Title	Type	Service Class	Size
1	B-01	Ticket search returns incorrect results for date-based searches	Bug	Standard	Small
2	B-02	Closed tickets appear in "Open Tickets" view	Bug	Standard	Small
3	B-03	Ticket comments display in wrong order	Bug	Standard	Small
4	B-04	Assign ticket doesn't validate staff name	Bug	Standard	Small
5	B-05	Search by ticket ID fails for single-digit IDs	Bug	Standard	Small
6	F-01	Add priority field to tickets	Feature	Standard	Medium
7	F-03	Add ticket categories	Feature	Standard	Medium
8	I-01	Refactor ticket listing for better performance	Improvement	Intangible	Medium
9	I-03	Add input validation and error handling	Improvement	Intangible	Medium

Total: 9 items · Standard: 7 · Intangible: 2

Goal for Lecture 9: Complete 5–7 items to demonstrate continuous flow